

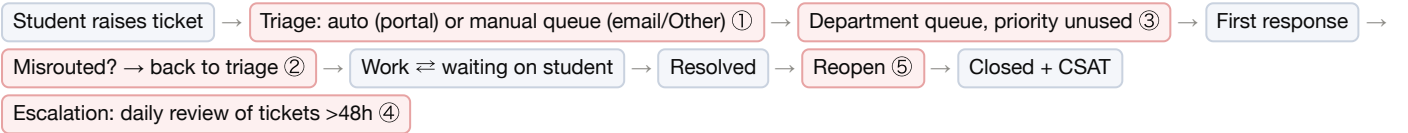
1 · One-page problem brief

Client: Student Services Helpdesk of a residential engineering college (about 1,800 students; 15 helpdesk staff across 5 departments plus triage). **Sponsor:** Dean of Student Affairs. **Original request:** "Buy an AI chatbot because tickets take forever." **FDE reframing:** find where the time actually goes first.

Stakeholders

Who	Role in problem	What they want
Dean of Student Affairs	Sponsor, owns the KPI	Fewer complaints; a defensible decision on the chatbot spend
Helpdesk Manager	Operational owner	Know which queue or step is slow; fair workload
Triage coordinators (2)	Route email/'Other' tickets	Less manual sorting
Department agents (IT, Hostel, Finance, Academics, Library)	Resolve tickets	Receive only tickets that belong to them
Students	Affected users	Fast, first-time-right resolution; visibility
IT admin	Owns ticketing tool data	No disruption to the tool

Current workflow (where the problem appears)



Before the problem: intake and categorisation decide the first queue. After the problem: students chase with follow-ups, reopen tickets and give low CSAT.

Problem statement (SMART)

Only **63%** of student helpdesk tickets resolved between 27 Jul and 20 Sep 2026 met their priority SLA (High 24h / Medium 72h / Low 120h), and only **21%** of High-priority tickets did. **Identify where in the ticket workflow delay accumulates** and implement the highest-impact process fixes to raise SLA compliance to **≥75%** (High-priority to **≥50%**) **within 8 weeks**, using the **existing helpdesk team and current ticketing tool** (no new hires, no new software).

KPI & supporting metrics

Metric	Baseline
KPI: SLA compliance rate (resolved within priority SLA, net of waiting-on-student)	63.1%
M1 Median time to first response	17.9h
M2 Median manual-triage wait	13.2h
M3 Reassignment rate	12.4%
M4 Reopen rate	9.4%
M5 Average CSAT (1–5), response rate	3.69 (36%)

Scope

MVP (in): all tickets from the 3 intake channels over 8 weeks; the lifecycle from creation to final resolution; routing, queueing, escalation and reminders; a repeatable pipeline and KPI report.

Out of scope: building or buying the chatbot; staffing and HR changes; agent-level performance ranking; real-time dashboards; changes to the ticketing tool's code.

Success means: the KPI is **≥75%** and High **≥50%** on the same, signed-off definition, with reassignment rate and median first response falling. Early read on leading indicators within 2 weeks of the pilot.

Data: synthetic, realistic (seeded generator). Sources: SQLite ticket DB, CSV audit log, JSON CSAT export, paginated Portal REST API.

2 · What I found

63.1%

SLA compliance (KPI), 1,069 resolved tickets · target 75%

21%

High-priority tickets within 24h

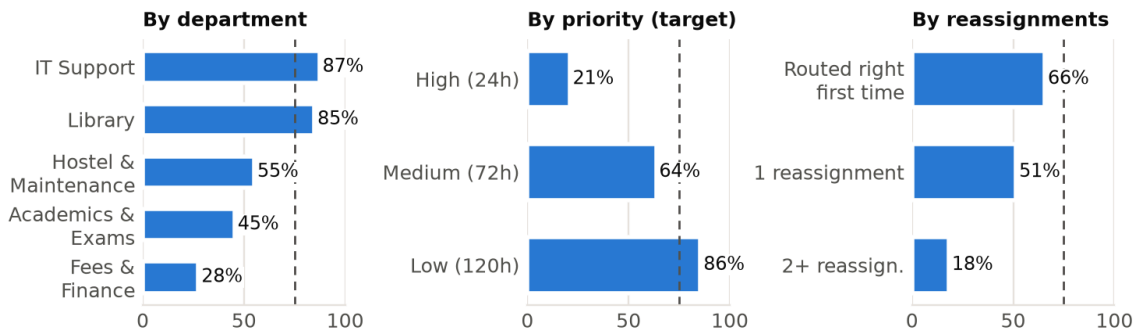
66→18%

SLA: routed right first time → 2+ reassignments

126h vs 42h

avg time, breached vs on-time tickets

SLA compliance, % of resolved tickets (dashed line = 75% target)



- **Priority is captured but ignored:** High tickets wait as long for a first response (19.3h median) as Low (16.3h).
 - **Misrouting compounds delay:** 12% of tickets are reassigned, and each hop cuts the SLA odds (66% → 51% → 18%).
 - **Two teams hold the backlog:** Fees & Finance 28% and Academics 45% (weekday office hours) vs IT 87%.
 - **Escalation is too late:** a daily review of tickets open >48h (median 65h); 132 tickets escalated only after breaching. Students aren't the cause (about 3h waiting on them either way).
- Recommendation:** don't buy the chatbot yet. Fix intake routing, escalate at 50% of each ticket's SLA, and sort queues by deadline. The estimate is about **79%** (above the 75% target). Pilot for 2 weeks with leading indicators (reassignment rate, triage wait, on-time escalations) before scaling.

3 · What I built (evidence in the repo)

Skill area	Evidence	Where
1 FDE mindset & stakeholders	Stakeholders, goals, constraints, facts / assumptions / unknowns; chatbot request reframed	README §1
2 Workflow & problem structure	Current-state workflow flowchart, 5 pain points, MECE issue tree, 5 hypotheses, SMART statement	README §2
3 KPIs, scope & solution	KPI = SLA compliance (63.1%), 5 supporting metrics, MVP / out-of-scope, success criteria	Page 1 · README §3
4 Client data & sources	Source map with source-of-truth choices, key fields, missing events; synthetic data generation logic	README §4 · source_systems/
5 Retrieval (4 types)	SQL (SQLite) · CSV audit log · nested JSON · paginated REST API with retries; completeness proven (4,550/4,550 records); raw snapshot + sha256 manifest	README §5 · retrieve.py
6 Profiling & cleaning	Profiling + 15 business rules → PASS/WARN/FAIL/UNKNOWN gate (6 PASS, 8 WARN, 1 UNKNOWN); safe fixes only, semantic issues flagged to owners	README §6 · validate.py
7 Workflow model	Entities, events, interactions, interventions, outcomes → ticket_journey (1 row/ticket); all metrics in SQL joins/aggregations	README §7 · model.py, metrics.py
8 Dependable pipeline	retrieve → validate → model → metrics → publish; per-run logs, exit codes, atomic publish; API outage → exit 2, truncated export → gate FAIL, nothing published; CI re-runs all on every push	README §8 · logs/ · CI

Reliability: against the generator's hidden ground truth (63.5%), the pipeline gives 63.1% (-0.4 pp, 99% per-ticket agreement). A naive query silently drops 42% of tickets.

4 · What remains unknown

- **Why** tickets are misrouted: there's no reassignment reason code, so instrument one.
- Causal effect of earlier escalation (only slow tickets get escalated, so the pilot must test it); real agent effort (no "start work" event); CSAT bias (36% response).
- The SLA definition isn't signed off by the Dean (the KPI is 57.5–65.9% across definitions, and the conclusion holds under all of them).

GitHub repo <https://github.com/LAKSHYAMEWARA0025/campusdesk-fde-assignment> (README explains every stage, with flowcharts)

Notebook · pipeline [walkthrough notebook](#) · [logs/](#) · [validation_report.md](#) · [CI runs](#) · [detailed report \(optional\)](#)

5-min demo video: [Google Drive link to be added](#)